

Entrepreneurial Marketing – Basic Analytics

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UNBOXING BASIC MARKETING ANALYTICS

CASE STUDY: DROPBOX

Imagine...

You are working in a new startup with an online-tool (SaaS).

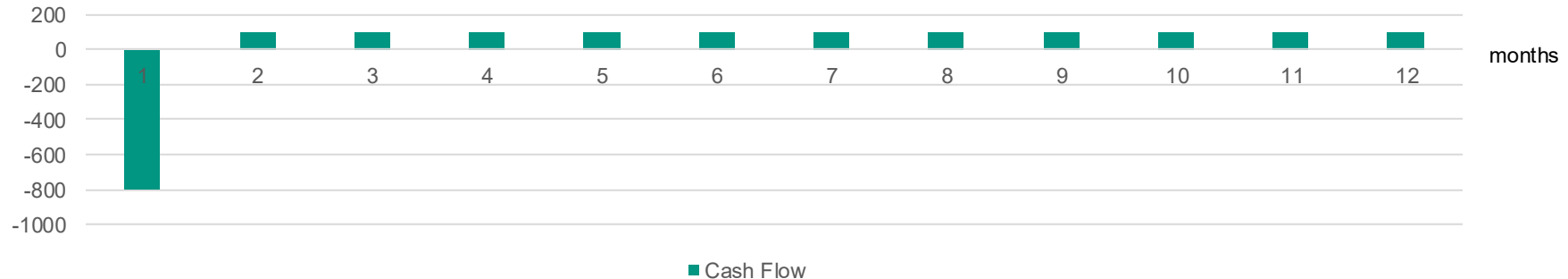
Customers can use it in a limited version for free and pay for the full range of functions.

What are the implications of this freemium business model?



The Challenge of Subscription Models

- SaaS and subscription businesses are front-loaded:
 - Companies invest upfront to acquire each customer.
 - But the revenue is deferred, coming in slowly over time.
- This creates a cash flow gap: High cost today vs. revenue tomorrow.



Three Key Metrics for Managing This Model

■ **LTV – Customer Lifetime Value**

How much gross profit does one customer generate over the lifetime?

Factors: monthly revenue, retention, churn, costs to serve.

■ **CAC – Customer Acquisition Cost**

How much does it cost to acquire one customer?

Includes marketing, sales, onboarding, etc.

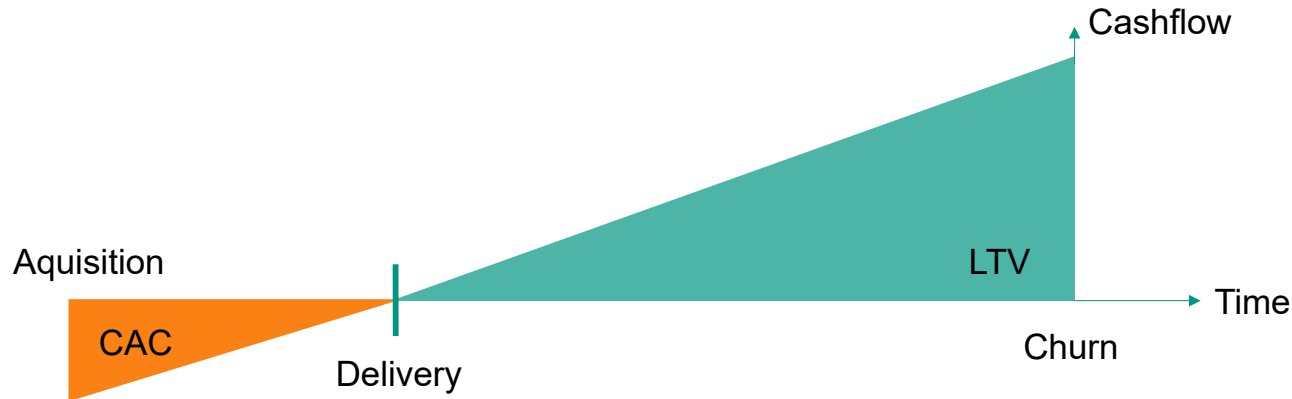
■ **NPV – Net Present Value**

What is a customer worth today, considering future value discounted over time?

Essential for comparing investments and managing capital.

Why These Metrics Matter

- They tell us whether growth is profitable.
- Golden rule: $LTV > CAC$ = sustainable business.
- $LTV - CAC$ Ratio matters.
- NPV helps decide: Is acquiring a customer creating long-term value?



Case Study: Dropbox Business Model

- Founded: 2007, by Drew Houston and Arash Ferdowsi
- Business model: Freemium (Free 2GB → Paid subscription for more storage)
- Goal: Easy, reliable file storage & sharing
- Customer focus: Individuals and small businesses



Recent Examples



Key Customer Metrics Dropbox in 2010

- 4 million users
- 3% Paying Customers (~120,000)
- Average Revenue per Paying User (ARPU): ~€125/year
- Retention Rate: 79%
- Churn Rate: 21% (100% - 79%)

User Type	Number of Users	Average Revenue per User	Annual Cost per User	Gross Margin per User
Free Users	3,880,000	\$0	\$1.32	-\$1.32
Paying Users	120,000	\$125	\$38.16	\$86.84

Churn Rate and Customer Lifetime

- The churn rate measures the percentage of customers who stop using a service during a specific time period.
- Customer Lifetime refers to the expected duration (usually in months or years) that a customer continues to use and pay a service before churning (leaving).
- Dropbox in 2010:
 - $Churn\ Rate = 1 - Retention\ Rate = 1 - 0.79 = 0.21$
 - $Customer\ Lifetime = \frac{1}{Churn\ Rate} = \frac{1}{0.21} \approx 5\ \text{years}$

Understanding Present Value and Discount Rate

Discount Rate

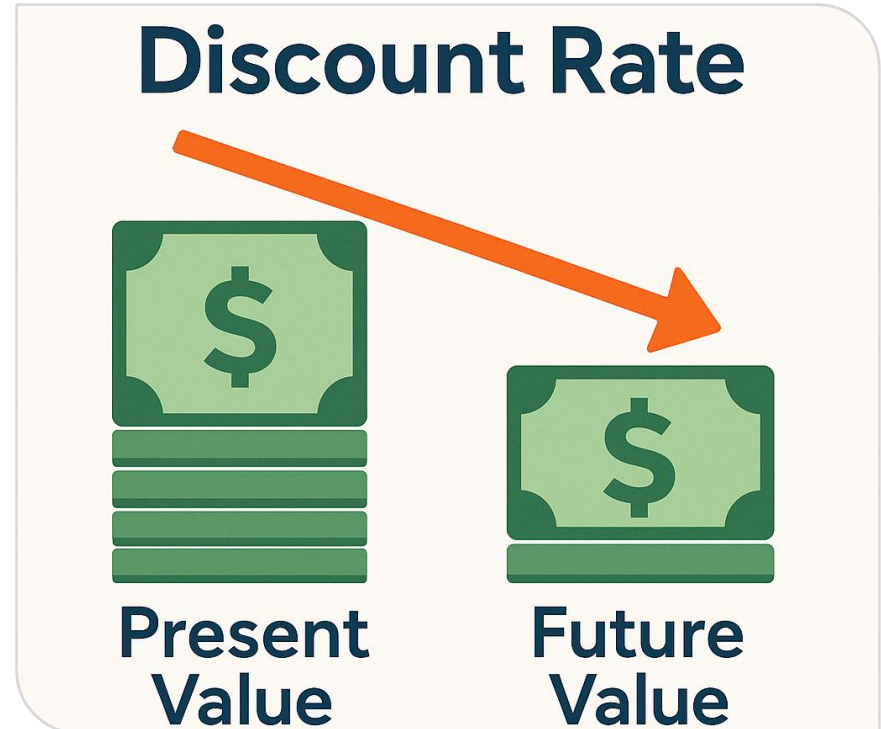
- The discount rate reflects the time value of money: Money today is worth more than money tomorrow
- It adjusts future cash flows to reflect their value today.
- Key reasons: investment opportunities, inflation, and risk.

Present Value

$$\text{Present Value} = \frac{\text{Future Cashflow}}{(1+r)^t}$$

r = discount rate

t = number of periods (years or months)



Customer Lifetime Value (LTV) Calculation

- Discount Rate: 15%
- Customer Lifetime: 5 years
- Variable Contribution = *Gross margin per user*
- Gross Margin per User = *Revenue per User – Cost per User*
- LTV per Customer: $\sum_{n=1}^N \frac{\text{Variable Contribution}_{\text{Year } n}}{(1 + \text{Discount Rate})^n}$

$$= \frac{\text{Variable Contribution}_{\text{Year } 1}}{(1+0,15)^1} + \frac{\text{Variable Contribution}_{\text{Year } 2}}{(1+0,15)^2} + \frac{\text{Variable Contribution}_{\text{Year } 3}}{(1+0,15)^3} + \dots + \frac{\text{Variable Contribution}_{\text{Year } N}}{(1+0,15)^N}$$

LTV Calculation for Dropbox's Paying Users

- Discount Rate: 15%
- Customer Lifetime: 5 years

$$LTV = \sum_{n=1}^5 \frac{\text{Variable Contribution}}{(1 + \text{Discount Rate})^n} = \$291.10$$

Year	Variable Contribution	Discount Rate	Present Value (PV)
1	\$86.84	15%	\$75.51
2	\$86.84	15%	\$65.66
3	\$86.84	15%	\$57.10
4	\$86.84	15%	\$49.65
5	\$86.84	15%	\$43.17
Total			\$291.10

Total LTV of 120,000 Paying Users: **\$34,932,137.85**

LTV Calculation for Dropbox's Free Users

- Discount Rate: 15%
- Customer Lifetime: 5 years

$$LTV = \sum_{n=1}^5 \frac{\text{Variable Contribution}}{(1 + \text{Discount Rate})^n} = -\$4.42$$

Year	Variable Contribution	Discount Rate	Present Value (PV)
1	-\$1.32	15%	-\$1.15
2	-\$1.32	15%	-\$1.00
3	-\$1.32	15%	-\$0.87
4	-\$1.32	15%	-\$0.75
5	-\$1.32	15%	-\$0.66
Total			-\$4.42

Total LTV of 3,880,000 Free Users: **-\$17,168,397.55**

Total LTV for All Customers

LTV per Pay-User: \$291.10

LTV per Free-User: -\$4.42

LTV for all Customers:

\$34,932,137.85 – \$17,168,397.55 = \$17,763,740.30

CAC (Customer Acquisition Cost)

- CAC = the present value of the costs of acquiring a customer.
- Acquisition costs can be:
 - Marketing efforts
 - Paid Ads
 - Promotional discounts (including costs of free users)
 - ...

Dropbox CAC Calculation

- 3 million **new users** were added from June 2009 to 2010
- PV of Acquisition Expenses:
 - Referral Program Cost (additional storage)¹
 - Free User Costs (storage)²
 - Google AdSense³
- Cost Calculation: $^163,000 + ^21,470,000 + ^33,500 = 1,536,500 \approx \$1,540,000$
- New Customers Acquired: 90,000
- $$CAC = \frac{PV \text{ of Acquisition Costs}}{New \text{ Customers Acquired}} = \frac{1,540,000}{90,000} = \$17.11 \text{ per customer}$$

NPV per Customer

Net Present Value (NPV) per customer = lifetime value (LTV) minus customer acquisition cost (CAC)

$$NPV = LTV - CAC = \$291.10 - \$17.11 = \$\mathbf{273.99}$$

NPV per customer is positive!

Dropbox growth strategy was profitable after referral pivot.

What if $CAC \geq LTV$?

- Overthink Business Model

or

- Lower CAC
- Improve LTV



How Dropbox Lowered CAC

1. Referral Program

("Give 250MB, Get 250MB")

- Incentivized existing users to invite friends.
- 35% of new users came from referrals.
- Extremely low cost compared to ads.

2. Viral Product Features

- File sharing and public links created natural viral loops.
- Sharing a folder often "forced" new users to sign up.

3. Product-Led Growth

("It Just Works")

- Superior ease-of-use and reliability.
- Happy users naturally recommended Dropbox.

4. Abandoning Paid Ads

- Early Google Ads cost > \$300 per paying customer.
- Dropbox stopped inefficient paid marketing.
- Focused entirely on organic growth.

How Dropbox Increased LTV

1. High Customer Retention

- Retention rate: 79%
- Longer customer relationships meant more payments.

2. Upselling Extra Storage & Features

- Free → Paid upgrades.
- Add-ons like "Pack-Rat" increased revenue per user.

3. Trojan Horse Business Strategy

- Individuals brought Dropbox into companies.
- Later, official enterprise adoption expanded LTV.

4. Low Cost-to-Serve (Efficient COGS)

- Used Amazon S3 for scalable, cheap storage.
- Smart policies (e.g., limited free undo history) reduced storage costs.

Key Learnings from Dropbox

- Freemium + Organic Growth → Sustainable scaling
- CAC has a huge impact on profitability: Referral System reduced CAC dramatically
- NPV calculation essential to understand profitability despite offering "free" service
- Investing early in user experience and viral mechanics pays off → higher LTV

“Freemium is a spreadsheet game – one you win with lots and lots of little moves” – Drew Houston

References Case Study

Dropbox: 'It Just Works'

By: Thomas R. Eisenmann, Michael Pao, Lauren Barley

Harvard Business School, 2011, 811065-PDF-ENG